Project 2: Linear Regression House Price Prediction – Task Documentation

1. Project Title

House Price Prediction using Linear Regression

2. Task Duration

05 March 2026 – 07 March 2026

3. Objective

The main objective of this project is to predict house prices using the Linear Regression machine learning algorithm. By analyzing historical housing data such as area, number of bedrooms, and location factors, the model estimates the expected price of a house.

4. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook / Google Colab

5. Dataset Description

The dataset contains information about houses such as:

- Area (square feet)
- Number of Bedrooms
- Number of Bathrooms
- Location factors
- Price of the house

This data is used to train the Linear Regression model to predict house prices.

6. Data Preprocessing

Before building the model, the following preprocessing steps were performed:

- Loaded the dataset using Pandas
- Checked for missing values
- Handled missing values using mean imputation
- Removed duplicate records
- Converted categorical values if required

7. Model Implementation

Steps used to implement Linear Regression:

1. Import required libraries
2. Load dataset
3. Separate independent variables (features) and dependent variable (price)
4. Split the dataset into training and testing sets
5. Train the Linear Regression model using Scikit-learn
6. Predict house prices using the trained model

8. Data Visualization

Visualization was performed using Matplotlib to understand relationships between variables:

- Area vs Price
- Bedrooms vs Price
- Predicted Price vs Actual Price

These graphs help in understanding housing price trends.

9. Results

The Linear Regression model successfully predicts house prices based on input features. The prediction results show a strong relationship between house area and price.

10. Conclusion

This project demonstrates how Linear Regression can be used to predict house prices based on available housing data. The model provides useful insights for real estate analysis.

11. Future Improvements

- Use larger datasets
- Include more features such as location and amenities
- Apply advanced machine learning models
- Improve accuracy using feature engineering